

Lesson 4 — Ansible Roles & Playbooks · Homework

Topics: role directory structure (`tasks/` , `handlers/` , `templates/` , `vars/`), `notify` /handlers, idempotency, `site.yaml` mapping roles to host groups, `--limit` , tags.

Use the `webserver` and `database` roles already in this folder as your reference — you're extending them, not starting from a blank page.

Part A — Read before you touch anything

Answer these in `answers.md` , in your own words (2–3 sentences each):

1. Why does `roles/webserver/tasks/main.yaml` get run automatically just because `site.yaml` says `roles: [webserver]` — what is Ansible actually doing to find it?
2. `handlers/main.yaml` has a `Restart nginx` handler. Under what condition does it actually run, and what would happen if you turned it into a normal task instead?
3. `vars/main.yaml` defines `web_packages` . Where in `tasks/main.yaml` is that variable used, and what happens if you add a third package to the list?

Part B — Extend the `webserver` role

1. Add a third package to `web_packages` in `roles/webserver/vars/main.yaml` (e.g. `http`).
2. Add a new task to `roles/webserver/tasks/main.yaml` that deploys a **second** template file of your choice (e.g. a custom `robots.txt` or a `motd` banner) — you'll need a new file under `templates/` .
3. Run `ansible-playbook site.yaml --limit web` and confirm only the web hosts were touched (the `db` play should show `0` tasks for its hosts).
4. Run the exact same command a **second time** and paste the output. Point at the `changed=0` line and explain in one sentence why nothing changed the second time.

Part C — Extend the `database` role

1. Add a task to `roles/database/tasks/main.yaml` that creates a database using the `mysql_db` module (name it anything, e.g. `app_db`).
2. Run `ansible-playbook site.yaml --limit db` and confirm it applies only to the `db` host(s).

Deliverables

- `answers.md` with Part A's answers and the command output pairs from Part B.3–B.4.
- Your edited `roles/webserver/vars/main.yaml`, the new template file, and the updated `roles/database/tasks/main.yaml`.

Bonus

- Scaffold a brand-new role with `ansible-galaxy init roles/monitoring` and compare its generated folders to `roles/webserver` — list which folders `webserver` doesn't use, and why that's fine.
- Add `tags: ['webserver']` / `tags: ['database']` to the two roles in `site.yaml` and run `ansible-playbook site.yaml --tags webserver` — confirm it behaves the same as `--limit web` did, and explain in 2 lines a situation where `--tags` can do something `--limit` can't.